Ved Ambarkar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.28
R.S. Chandak High School and S.E.S. Jr. College, Solapur	Class XII (MSBSHSE)	2024	89.67%
KLE English Medium School, Jule Solapur	Class X (CBSE)	2022	94.80%

SCHOLASTIC ACHIEVEMENTS

Rank 1, Optiver Trade-a-thon 2025, IITH: Competed in a live-trading simulation, utilizing mathematical expressions and rapid mental math to predict future asset prices, execute profitable trades, and manage risk against other participants.

Centre Topper (Rank 1), National Standard Examinations in Physics (NSEP) & Astronomy (NSEA) 2023.

SKILLS

Programming Languages: C++, Python, C, Bash, LaTeX

AI & Scientific Computing: PyTorch, TensorFlow, Keras, NumPy, SciPy, Pandas, Custom Inference Architectures

Systems & Low-Latency: Unix/POSIX APIs (kqueue, libproc), Memory Management (Zero-Heap Allocation, Cache Locality), Multithreading, Event-Driven Architecture, Operating Systems Scheduling

Developer Tooling: Git, CMake, LLDB/GDB (or System Profilers), Shell Scripting, Linux/Darwin Environments

PROJECTS

Post-Training Quantization & KV-Cache Transformer Optimizer | GitHub Link

Tech: Python, PyTorch, NumPy, Transformer Architectures, Model Quantization, LLM Optimization

- Compressed FP32/FP16 weight matrices down to **INT8 precision** through a **post-training quantization (PTQ)** pipeline using linear scaling and zero-point calibration, shrinking memory footprint by ~50%.
- Replaced redundant historical token re-computations by integrating a dynamic **Key-Value (KV) Cache** into multi-head attention blocks, cutting autoregressive decoding complexity from **O(N^2) down to O(N)**.
- Guided generation randomness and output stability by implementing **Top-K, Top-P (Nucleus), and Temperature scaling** sampling algorithms, unlocking **3x faster token throughput** with negligible perplexity loss.

Custom C++ Deep Learning Inference Engine | GitHub Link

Tech: C++, Linear Algebra, Matrix Vectorization, Memory Locality, High-Performance Computing

- Wrote a standalone deep learning inference engine in pure C++, running forward-pass neural network calculations entirely from scratch without heavyweight wrappers like PyTorch.
- Hand-crafted custom **matrix multiplication kernels** explicitly optimized for **cache-line locality** and **CPU SIMD vectorization**, drastically speeding up Linear, Softmax, and Convolutional layers.
- Guaranteed numerical stability across complex multidimensional array transformations by enforcing strict **contiguous memory contracts**, keeping data throughput at the absolute hardware limit.

High-Performance Quantitative Trading Engine | GitHub Link

Tech: C++, STL, Market Microstructure, Event-Driven Simulation, Low-Latency Systems

- Built a **deterministic limit order book** from scratch in C++, relying on **custom memory pools** and **intrusive linked**

lists to strip out **OS heap allocation overhead** during critical matching cycles.
- Achieved **sub-microsecond latency** across an **event-driven execution pipeline** by organizing trade data into **dense, flat-array memory layouts** that prevent CPU cache misses.
- Stress-tested order matching against simulated high-volume market bursts, isolating outliers to guarantee **deterministic throughput** regardless of order-book depth.

macOS Real-Time Process Telemetry & Lifecycle Monitor | GitHub Link

Tech: C++, Objective-C++, macOS Kernel APIs (kqueue, libproc), CoreAudio, IOKit, XNU Internals

- Mapped live process hierarchies and application lifecycles by leveraging **kqueue and libproc**, dynamically resolving background PID families without relying on inefficient polling loops.
- Probed Apple's **CoreAudio and IOKit** kernel subsystems through a custom Objective-C++ bridge, harvesting real-time **per-process CPU/GPU utilization** and hardware state metrics.
- Bypassed user-space abstraction layers in favor of native **XNU kernel primitives**, streaming telemetry through a **non-blocking ingestion pipeline** designed for **Apple Silicon (P-core/E-core) QoS scheduling**.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Applications, Operating Systems I, Operating Systems II

Computational Methods: Scientific Computing, Computer-Aided Numerical Methods I, Computer-Aided Numerical Methods II

Mathematics: Calculus, Linear Algebra, Differential Equations, Transform Techniques, Probability, Statistics, Complex Variables

EXTRACURRICULAR

Competitive Programming: Attained a Codeforces rating of **1400+ (Specialist tier),** consistently participating in global algorithmic contests.

POSITIONS OF RESPONSIBILITY

Core

Core in the Competitive Programming Club of IIT Hyderabad

Core

Core in the Books and Movies Club at IIT Hyderabad